HW #5 Problem # 1

Rework Murphy 2.67 with Oyster shell composition
2.1 wt% A, 55 wt% B and 42.9 wt% I

For the six species (A, B, C, D, E, and I) we can write six mass balances.

$$A: \quad 0.01\,m_1 + 0.68\,m_2 + 0.21\,m_4 = 21 \text{ tons}$$

$$B: \quad 0.54\,m_1 + 0.55\,m_4 = 65 \text{ tons}$$

$$C: \quad 0.20\,m_2 = 100\,w_{6C}$$

$$D: \quad 0.4\,m_2 + m_3 = 100\,w_{6D}$$

$$E: \quad 0.04\,m_1 = 100\,w_{6E}$$

$$I: \quad 0.41\,m_1 + 0.08\,m_2 + 0.429\,m_4 = m_5$$

This gives 6 eqns and 7 unknowns ($m_1, m_2, m_3, m_4, m_5, w_{6C}, w_{6E}$)
We will set the value of w_{6D} between 0.01 and 0.05.
We need one additional equation.

$$w_{6A} + w_{6B} + w_{6C} + w_{6D} + w_{6E} = 1$$

$$0.21 + 0.65 + w_{6C} + w_{6D} + w_{6E} = 1$$

$$w_{6C} + w_{6E} = 1 - 0.21 - 0.65 - w_{6D}$$

This is the 7th equation $\Rightarrow$ $w_{6C} + w_{6E} = 0.14 - w_{6D}$

Now write the 7 eqns in matrix form.

$$\begin{array}{ccc} A & X & = B \end{array}$$

$$\begin{pmatrix} 0.01 & 0.68 & 0 & 0.21 & 0 & 0 & 0 \\ 0.54 & 0 & 0 & 0.55 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 & -100 & 0 \\ 0 & 0.04 & 1 & 0 & 0 & 0 & 0 \\ 0.04 & 0 & 0 & 0 & 0 & 0 & -100 \\ 0.41 & 0.08 & 0 & 0.429 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \\ w_{6C} \\ w_{6E} \end{pmatrix} = \begin{pmatrix} 21 \\ 65 \\ 0 \\ 100 w_{6D} \\ 0 \\ 0 \\ 0.14 - w_{6D} \end{pmatrix}$$

Pick a value of w_{6D} and solve by taking A^{-1} and multiplying by B. This can be done numerically using Excel, MatLab, etc. I used Live Math (see next page.)

The calculate cost. Again $w_{6D} = 0.05$ gives lowest cost,

$$\begin{pmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \\ w_{6C} \\ w_{6D} \end{pmatrix} = \begin{pmatrix} 82.403 \\ 28.519 \\ 3.8592 \\ 37.277 \\ 52.659 \\ 0.057039 \\ 0.032961 \end{pmatrix} \quad \text{when } w_{6D} = 0.05$$

COST = \$10,486 minimum in range $0.01 \leq w_{6D} < 0.05$

$$A = \begin{pmatrix} 0.01 & 0.68 & 0 & 0.021 & 0 & 0 & 0 \\ 0.54 & 0 & 0 & 0.55 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 & -100 & 0 \\ 0 & 0.04 & 1 & 0 & 0 & 0 & 0 \\ 0.04 & 0 & 0 & 0 & 0 & 0 & -100 \\ 0.41 & 0.08 & 0 & 0.429 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 21 \\ 65 \\ 0 \\ 100\, wD \\ 0 \\ 0 \\ 0.14 - wD \end{pmatrix}$$

$$\begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix} = A^{-1} B$$

$$\begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix} = \begin{pmatrix} 0.01 & 0.68 & 0 & 0.021 & 0 & 0 & 0 \\ 0.54 & 0 & 0 & 0.55 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 & -100 & 0 \\ 0 & 0.04 & 1 & 0 & 0 & 0 & 0 \\ 0.04 & 0 & 0 & 0 & 0 & 0 & -100 \\ 0.41 & 0.08 & 0 & 0.429 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 21 \\ 65 \\ 0 \\ 100\, wD \\ 0 \\ 0 \\ -wD + 0.14 \end{pmatrix}$$ Substitute

$$\begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix} = \begin{pmatrix} 0.01 & 0.68 & 0 & 0.021 & 0 & 0 & 0 \\ 0.54 & 0 & 0 & 0.55 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 & 0 & -100 & 0 \\ 0 & 0.04 & 1 & 0 & 0 & 0 & 0 \\ 0.04 & 0 & 0 & 0 & 0 & 0 & -100 \\ 0.41 & 0.08 & 0 & 0.429 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 21 \\ 65 \\ 0 \\ 5 \\ 0 \\ 0 \\ 0.09 \end{pmatrix}$$ Substitute

$$\begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix} = -0.62004 \begin{pmatrix} 11 & -0.42 & -37.4 & 0 & -37.4 & 0 & -3740 \\ -2.2 & 0.084 & -0.584 & 0 & -0.584 & 0 & -58.4 \\ 0.088 & -0.00336 & 0.02336 & -1.6128 & 0.02336 & 0 & 2.336 \\ -10.8 & -2.52 & 36.72 & 0 & 36.72 & 0 & 3672 \\ -0.2992 & -1.2466 & 0.37216 & 0 & 0.37216 & 1.6128 & 37.216 \\ -0.0044 & 0.000168 & 0.01496 & 0 & -0.001168 & 0 & -0.1168 \\ 0.0044 & -0.000168 & -0.01496 & 0 & 0.001168 & 0 & -1.496 \end{pmatrix} \begin{pmatrix} 21 \\ 65 \\ 0 \\ 5 \\ 0 \\ 0 \\ 0.09 \end{pmatrix}$$ Expand

$$\begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix} = \begin{pmatrix} 82.403 \\ 28.519 \\ 3.8592 \\ 37.277 \\ 52.059 \\ 0.057039 \\ 0.032961 \end{pmatrix}$$ Expand

$$wD = 0.05$$

$$COST = (88, 35, 0, 60, 0, 0, 0) \begin{pmatrix} ma \\ mb \\ mc \\ md \\ wC \\ wE \end{pmatrix}$$

$$COST = (88, 35, 0, 60, 0, 0, 0) \begin{pmatrix} 82.403 \\ 28.519 \\ 3.8592 \\ 37.277 \\ 52.059 \\ 0.057039 \\ 0.032961 \end{pmatrix}$$ Substitute

$$COST = 10486$$ Expand